Functional Environmental Flows from Reinforcement Learning: a Proof of Concept on a Lake McClure–like Reservoir with OpenHydroQual as the Simulator

OpenHydroQual-RL working notes

November 1, 2026

Abstract

Environmental flow (e-flow) requirements below dams are usually *prescriptive*: a fixed release schedule, chosen in advance for each water-year type. This note tests the alternative proposed in the "DRL for e-flows" outline: put the ecological objectives into the *reward* of a reinforcement-learning (RL) agent and let the agent decide, day by day, how to release water. We build a reservoir patterned on Lake McClure on the Merced River (California), drive it with the observed daily unimpaired inflow for water years 1989–2025, and simulate it with OpenHydroQual (OHQ) through its Python bindings. Each day the agent chooses the river release and the irrigation delivery. The ecological objective follows the five functional-flow components of the California Environmental Flows Framework (fall pulse, wet-season baseflow, peak flows, spring pulse and recession, dry-season baseflow). The targets are derived from the river's own unimpaired hydrology, separately for dry, moderate and wet years. We compare four operating modes on the 2017–2025 test period, which spans a record wet year, a drought and another wet year: (1) a regulatory minimum instream flow, (2) a prescribed functional-flow hydrograph, (3) functional flows in the reward, and (4) functional flows plus hydropower in the reward. Prescribing functional flows raises the functional-flow score from 0.57 (minimum flow) to 0.93, above the unimpaired river's 0.88. Overall irrigation delivery falls only from 99.9% to 97.3%, but the cost is concentrated in the third year of the 2020–22 drought (85%). A PPO agent with functional flows in its reward, started from the prescriptive rule, did *not* improve on it: every update made the noise-free policy worse. We document why, and the reward-design lessons learned on the way, which are themselves useful for the planned paper.

Contents

1 Motivation and what this note does **3**

2 The system **3**
2.1 A Lake McClure–like reservoir 3
2.2 Data 3
2.3 Parameters, demand, flood control and minimum flow 4

3 Functional flows as numbers **4**
3.1 The five components 4
3.2 Targets from the river's own natural flow 6
3.3 The prescribed hydrograph (Scenario 2) 7
3.4 Year type under uncertainty 8

4 The reinforcement-learning problem **8**
4.1 In one paragraph . . . 8
4.2 What the agent sees (state) . . . 8
4.3 What the agent decides (actions) . . . 8
4.4 What the agent is rewarded for . . . 9
4.5 Episodes, discounting and the end of an episode . . . 10

5 The simulator: one OHQ run per day **10**

6 How the agent is trained **10**

7 Baselines **11**

8 Results **11**
8.1 Summary of the operating modes . . . 12
8.2 Year by year . . . 12
8.3 Hydrographs . . . 13
8.4 A check against observed operations . . . 13
8.5 Reinforcement learning (Scenarios 3 and 4) . . . 13

9 Discussion and next steps **18**

1 Motivation and what this note does

Most e-flow requirements are *static*. A schedule of minimum flows, sometimes with pulses, is fixed in a licence and indexed to the water-year type. Static rules are easy to write down and to enforce, but they ignore information that operators actually have: the state of the reservoir, the snowpack forecast, and whether a natural storm is passing anyway. Under climate change the hydrology that a static schedule was designed for also drifts away from the hydrology it has to work with.

An *adaptive* rule maps what the operator knows today to a release today. Reinforcement learning is a general way to find such a rule by trial and error in a simulator. We specify *what* we want (the reward) and the algorithm searches for *how* to operate. The question the outline poses is whether e-flows can be expressed as part of that reward, so that an e-flow-friendly operation emerges from the optimization instead of being prescribed. The outline's Scenario 3, "reward-function functional flows", is its core novelty.

This note is a working proof of concept of that idea. It follows the outline's four scenarios on a single reservoir:

1. **Minimum instream flow** (MIF). A regulatory floor on the river flow; the agent (or rule) manages supply and floods above it.
2. **Prescribed functional flows.** A fixed functional-flow hydrograph for each year type replaces the MIF floor.
3. **Functional flows in the reward.** No floor. Every functional-flow component is scored daily and added to the reward.
4. **Multi-objective.** Scenario 3 plus hydropower in the reward (flood control is in every scenario).

The general RL background (states, actions, rewards, value functions, the PPO algorithm) is explained step by step in the companion tutorial report *Learning Reservoir Operating Rules with Reinforcement Learning* (`report/main.pdf`). Here we only summarize what is needed (Section 4) and concentrate on the e-flow problem.

2 The system

2.1 A Lake McClure–like reservoir

Lake McClure (New Exchequer Dam) on the Merced River stores about 1.02 million acre-feet (1264 hm^3). It is operated by the Merced Irrigation District for irrigation, hydropower and flood control, under FERC and other in-stream flow requirements. We do *not* attempt a faithful model of the real system (no McSwain/Merced Falls re-regulation, no evaporation, simplified head and flood rules). The aim is a stylized system with realistic hydrology, demand and scale, so that the trade-offs are real ones. Figure 1 shows the model.

2.2 Data

All data are public daily series (Table 1), fetched by `prepare_merced.py`.
Water years (WY) run October 1 to September 30. Leap days are dropped, so every year has 365 daily steps, with day $d = 0$ on October 1. The mean unimpaired runoff of the training years (WY1989–2016) is 1093 hm^3/yr and the irrigation demand is 644 hm^3/yr, so in an average year a little over half of the inflow is wanted for irrigation, and in a dry year all of it and more (Figure 2).

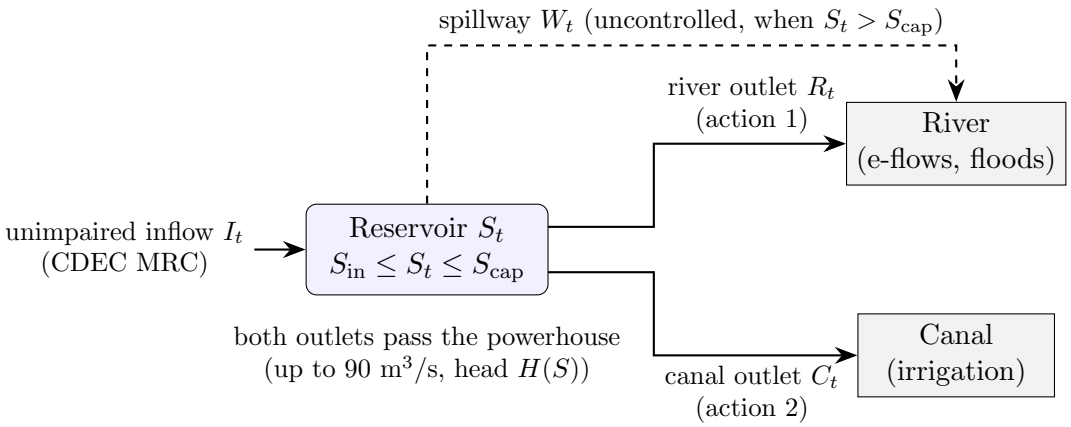


Figure 1: The model. Every day the agent sets the river release R_t and the canal delivery C_t. The spillway is uncontrolled. In OHQ the reservoir is a `Reservoir` block, the three receivers are `fixed_head` blocks whose storage accumulates the delivered volume, and the outlets are `Reservoir_link_rule` links.

Series	Source	Use	Period
Full natural (unimpaired) flow, Merced R. nr Merced Falls	CDEC MRC, sensor 8	inflow I_t, targets	WY1989–2025
Lake McClure storage	CDEC EXC, sensor 15	initial storage of test windows	1988–2025
Lake McClure outflow	CDEC EXC, sensor 23	irrigation demand	1994–2025
Merced R. near Snelling (below the diversions)	USGS 11272500	demand; historical river flow	WY1989–2010

Table 1: Data. The irrigation demand is the observed diversion, i.e. reservoir outflow minus the river flow below the Crocker–Huffman diversion dam (WY1994–2010).

2.3 Parameters, demand, flood control and minimum flow

The **minimum instream flow** of Scenario 1 is stylized after the FERC licence and the Davis–Grunsky requirements on the Merced: 75 cfs in late October, 180 cfs from November to March, 75 cfs in April and May, and 25 cfs from June to mid-October, with 60% of these in dry years. Figure 4 shows it in blue.

3 Functional flows as numbers

3.1 The five components

The California Environmental Flows Framework (CEFF; Yarnell et al. 2020) describes a healthy hydrograph of a California river by five *functional-flow components*, each tied to ecological functions:

Fall pulse (FA). The first rains of autumn. It flushes fine sediment and cues salmon migration and spawning.

Wet-season baseflow (WB). Sustained winter flow. It keeps habitat wetted for rearing.

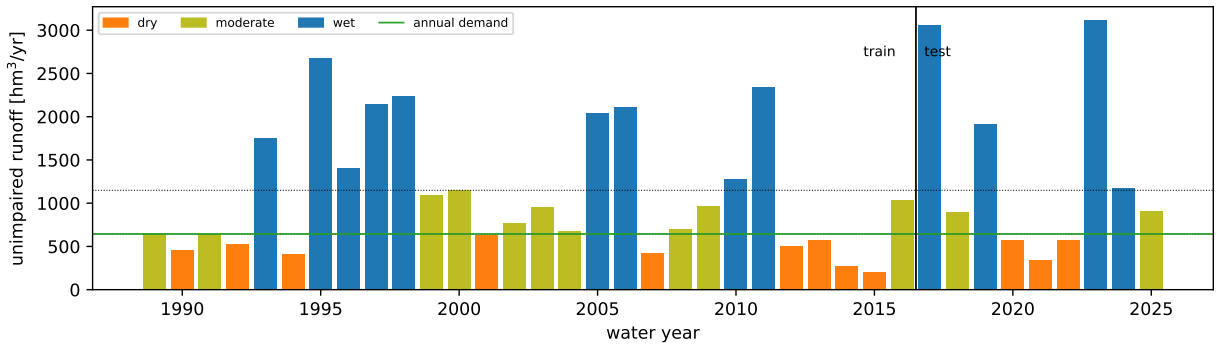


Figure 2: Annual unimpaired runoff and year types. The dotted lines are the terciles of the training years (640 and 1149 hm^3), which define dry, moderate and wet years. The green line is the annual irrigation demand. The test period WY2017–2025 contains the wettest year of the record (2023), the 2020–2022 drought and the wet 2017.

Symbol	Value	Meaning
S_{cap}	1264 hm^3	capacity (1.02 MAF); spillway opens above it
S_{in}	80 hm^3	inactive pool; both outlets taper to zero just above it (the observed minimum was 78 hm^3 in 2015)
$R_{\max}$	300 m^3/s	river outlet capacity
Q_{safe}	170 m^3/s	safe channel capacity (6000 cfs)
D_d	Fig. 3	irrigation demand on water-year day d; peak $D_{\max} = 50$ m^3/s
$S_{\mathrm{toc}}(d)$	Fig. 3	flood-control rule curve: 432 hm^3 (350 kAF) of flood space Nov 1–Mar 1
$H(S)$	70–140 m	gross head, $70+70\,x^{0.6}$ with $x = (S-S_{\mathrm{in}})/(S_{\mathrm{cap}}-S_{\mathrm{in}})$
P	$\leq$ 94.5 MW	$P = \rho g\,\eta\,\min(R+C,\,90)\,H$, $\eta = 0.88$

Table 2: System parameters.

Peak flows (PK). Large, infrequent floods. They move the bed, scour vegetation and connect the floodplain.

Spring pulse and recession (SP). The snowmelt pulse and its gradual decline. Spawning and seed dispersal follow it, and fish are stranded if it stops abruptly.

Dry-season baseflow (DS). Low summer flow. It sets the summer habitat and water temperature.

For an RL agent, "a healthy hydrograph" must become a number it can earn every day. We give each component a calendar window, a target, and a score. CEFF detects the timing of each component from the hydrograph; we use fixed windows to keep the rules readable (Table 3).
Two choices make the score robust to small release errors. Graded credit means a release 2% short of a target loses a little, not a whole day. Measuring the recession on the trailing 7-day mean flow, $\rho_d = (\bar{Q}_{d-1} - \bar{Q}_d)/\bar{Q}_{d-1}$, matches how $r_{\max}$ is derived from the natural flow (below). A first version with pass/fail days and raw daily declines turned out to be so sensitive that day-to-day release jitter of a few percent failed most of the spring (Section 6).

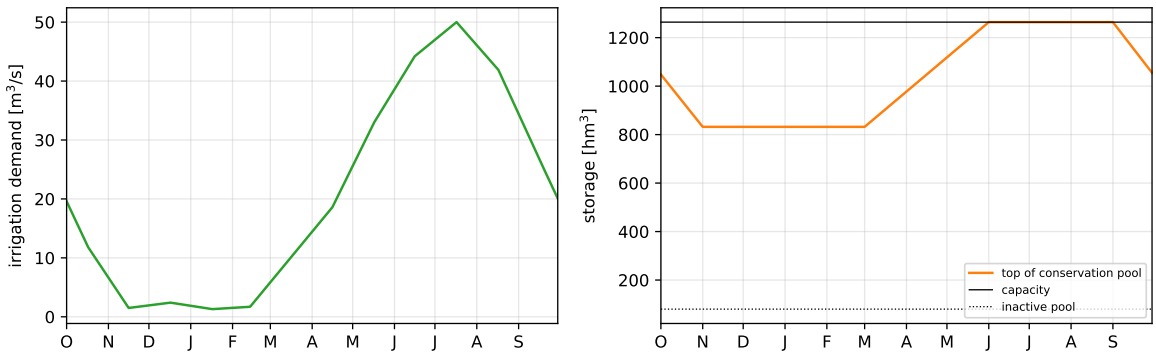


Figure 3: Left: daily irrigation demand (mean observed diversion per month, interpolated). Right: flood-control rule curve used by the rule-based baselines and given to the agent as an observation.

Component	Window	Requirement	Year score $\in [0,1]$
FA	Oct 15 – Dec 15	$\geq D_{FA}$ days with $Q \geq Q_{FA}$	$\min(1,\ \sum_d c_d / D_{FA})$
WB	Dec 16 – Mar 31	$Q \geq Q_{WB}$ every day	mean of c_d
PK	Dec 1 – Apr 30	$\geq D_{PK}$ days with $Q \geq Q_{PK}$	$\min(1,\ \sum_d c_d / D_{PK})$
SP	Apr 1 – May 31	$\geq D_{SP}$ days with $Q \geq Q_{SP}$	$\frac{1}{2}\min(1, \sum_d c_d / D_{SP})$
	Apr 15 – Jun 30	decline of the 7-day mean flow $\rho_d \leq r_{\max}$	$+\frac{1}{2}$ mean recession credit
DS	Jul 1 – Oct 14	$Q \geq Q_{DS}$ every day	mean of c_d

Table 3: Functional-flow components as used here (Q is the river flow below the reservoir, including spill). Day credit $c_d = \mathrm{clip}\big((Q_d/Q_k - 0.9)/0.09,\ 0,\ 1\big)$: full at 99% of the target or more, falling linearly to zero at 90%. Recession credit is 1 for $\rho_d \leq r_{\max}$ and falls linearly to 0 at $\rho_d = 2r_{\max}$.

The **functional-flow (FF) score** of a year is the mean of the component scores that apply. A pulse with a required duration of zero (peak flows in dry years) does not apply. A score of 1 means every component is fully met.

3.2 Targets from the river's own natural flow

Following CEFF, targets come from the unimpaired hydrology of the same river over the training years, for each year type:

- Q_{WB}: median over the years of the window's 10th-percentile flow (CEFF `Wet_BFL_Mag_10`);
- Q_{DS}: 25th percentile over the years of the window's median flow (the lower end of the natural range of `DS_Mag_50`);
- Q_{FA}: 25th percentile of the natural fall maxima, $D_{FA} = 2$ days;
- Q_{SP}: half the median natural spring maximum; D_{SP} the median number of natural days above it, capped at 7;
- Q_{PK}: the median annual peak (about the 2-year flood) capped at $0.8\,Q_{\text{safe}}$, because a regulated river cannot be asked to flood its own channel; D_{PK} is the median number of natural days above it (capped at 7);

- $r_{\max}$: 90th percentile of the natural daily declines of the 7-day mean flow during the recession window (about 10%/day, close to the CEFF guidance for Sierra snowmelt rivers).

Table 4 gives the values and Figure 4 shows them against the natural hydrographs. The unimpaired flow itself scores only 0.76–0.83, because natural years scatter around the targets: a natural dry year often has no fall pulse. A regulated river that scores near 0.8 is therefore about as "functional" as the natural one *by these metrics*.

Target	Unit	dry	moderate	wet
Fall pulse magnitude Q_{FA}	m^3/s	10.96	10.24	31.38
Fall pulse duration D_{FA}	d	2	2	2
Wet-season baseflow Q_{WB}	m^3/s	2.80	7.54	10.77
Peak flow magnitude Q_{PK}	m^3/s	136.00	136.00	136.00
Peak flow duration D_{PK}	d	0	1	7
Spring pulse magnitude Q_{SP}	m^3/s	44.64	75.23	120.25
Spring pulse duration D_{SP}	d	7	7	7
Max. recession rate $r_{\max}$	1/d	0.10	0.10	0.10
Dry-season baseflow Q_{DS}	m^3/s	0.34	1.67	4.11
Prescribed hydrograph volume	hm^3/yr	109	240	454
Minimum instream flow volume	hm^3/yr	53	89	89
FF score of the unimpaired flow	–	0.76	0.78	0.83

Table 4: Functional-flow targets by year type (from WY1989–2016 unimpaired flow), volume of the prescribed hydrograph and of the minimum instream flow, and the score that the unimpaired flow itself gets.

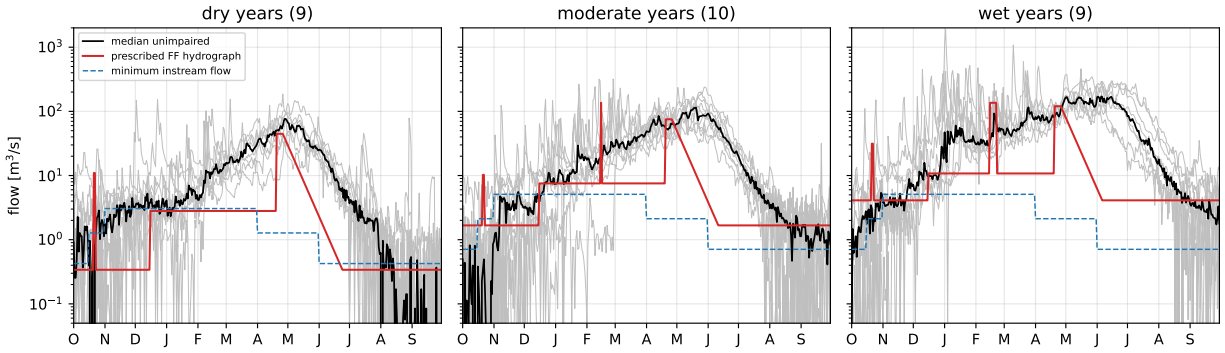


Figure 4: Unimpaired daily flow of the training years (gray; median in black), the prescribed functional-flow hydrograph used in Scenario 2 (red), and the minimum instream flow of Scenario 1 (blue, dashed). Log scale.

3.3 The prescribed hydrograph (Scenario 2)

The prescriptive alternative turns the targets into a fixed release schedule for each year type, met by construction: baseflow floors in their windows, a fall pulse on October 21, a peak-flow block on February 15 (moderate and wet years), and a spring pulse on April 20 followed by a recession at

$0.8\,r_{\max}$ down to the dry-season floor (red lines in Figure 4). It costs 109/240/454 hm^3 of river water in dry/moderate/wet years, against 53–89 hm^3 for the minimum instream flow.

3.4 Year type under uncertainty

Targets depend on the year type, but the type of the current year is not known until it is over. As in California practice, the operator works with a *declared* type based on the runoff forecast. We imitate a Bulletin-120 forecast: the remaining runoff of the water year is known with a log-normal error. The error's spread falls from 50% before February to 15% in April and 5% in June, and the error is redrawn every month. On each day the declared type is the tercile class of (runoff so far + forecast of the rest). Daily targets follow that day's declaration, and the year's score uses the May 1 declaration. A prescribed schedule therefore sometimes releases a "dry" winter baseflow in what turns out to be a moderate year, which is the real cost of deciding early.

4 The reinforcement-learning problem

4.1 In one paragraph

An RL *agent* (a small neural network, the *policy*) looks at the *state* of the system each day and chooses an *action*. The simulator advances one day and returns a *reward*. Training adjusts the network so that the expected sum of discounted rewards, $\mathbb{E}\big[\sum_t \gamma^t r_t\big]$, is as large as possible. Nothing about how to operate is programmed in. Everything the agent learns comes from the reward and the simulated consequences of its own choices. The companion report explains this in detail.

4.2 What the agent sees (state)

The observation o_t has 18 entries, all scaled to order one: storage S_t/S_{cap}; today's inflow and its 7- and 30-day means (as $\log(1+I/10)$); the season (sin, cos of $2\pi d/365$); the forecast of the remaining water-year runoff/S_{cap}; yesterday's river flow; today's demand/$D_{\max}$; today's *reference* river flow (in action units: the hard floor in Scenarios 1–2, the prescribed functional-flow schedule of the declared year type in Scenarios 3–4, where it is only a hint the agent may ignore); the declared year type (three 0/1 flags); the progress of the fall, peak and spring pulses this year (days achieved/days required); the flood space $(S_{\text{toc}} - S_t)/S_{\text{cap}}$; and a spill-risk indicator $(S_t + \text{forecast} - S_{\text{cap}})/S_{\text{cap}}$. The pulse-progress entries are needed because the pulse rewards depend on what has already been achieved this year. Without them the reward would not be a function of the observed state.

4.3 What the agent decides (actions)

Two numbers $a_1, a_2 \in [-1, 1]$:

$$R_t = R_{\min}\left(\frac{R_{\max}}{R_{\min}}\right)^{(a_1+1)/2}, \qquad C_t = D_d\,\min\big(1,\ 0.6\,(a_2+1)\big), \tag{1}$$

with $R_{\min} = 0.1$ and $R_{\max} = 300$ m^3/s. The river release is on a log scale because the targets range from 0.3 m^3/s (dry-season baseflow) to 136 m^3/s (peak flow): an error in a_1 is then the same *relative* flow error at every level. An earlier version with $R_t \propto (a_1+1)^2$ could not hold small baseflows accurately enough. Delivery is a fraction of today's demand, reaching 100% at $a_2 \geq 2/3$. In Scenarios 1 and 2 the river release is raised to the floor if needed, $R_t \leftarrow \max(R_t, F_t)$. Both outlets taper to zero just above the inactive pool, so the reservoir cannot be overdrawn.

4.4 What the agent is rewarded for

With $Q_t = R_t + W_t$ the river flow (release plus spill), the daily reward is the sum of these terms:

$$r_t^{\text{sup}} = -w_s \left(\frac{(D_d - C_t)^+}{D_{\max}} \right)^2 \qquad \text{irrigation shortage (squared)} \tag{2}$$

$$r_t^{\text{fl}} = -w_f \, \frac{(Q_t - Q_{\text{safe}})^+}{Q_{\text{safe}}} \qquad \text{flooding below the dam} \tag{3}$$

$$\begin{aligned} r_t^{\text{ff}} = {} & -w_{ff} \sum_{k \in \{WB,DS\}} \mathbb{1}_k(d) \, \frac{1 - c_t^{(k)}}{N_k} && \text{baseflow days missed} \\ & + \sum_{k \in \{FA,PK,SP\}} \omega_k \, w_{ff} \, \mathbb{1}_k(d) \, \Delta\min\Big(1, \frac{n_k}{D_k}\Big) && \text{pulse progress, } n_k \leftarrow n_k + c_t^{(k)} \\ & - \tfrac{1}{2} w_{ff} \, \mathbb{1}_{REC}(d) \, \frac{1}{N_{REC}} \min\left(1, \frac{(\rho_t - r_{\max})^+}{r_{\max}}\right) && \text{too-fast recession of the 7-day mean} \end{aligned} \tag{4}$$

$$r_t^{\text{hp}} = w_h \, P_t / P_{\max} \qquad \text{energy} \tag{5}$$

where $\mathbb{1}_k(d)$ is one inside the window of component k, N_k is the number of days in that window, $c_t^{(k)}$ is the day credit of Table 3, $\omega_{FA} = \omega_{PK} = 1$, $\omega_{SP} = \frac{1}{2}$, and n_k accumulates the day credits of pulse k this year.

The functional-flow reward is the score, paid daily. Summed over a year, r^{ff} equals w_{ff} times the sum of the component scores, minus a constant. Every component is worth $w_{ff} = 5$ per year, and the agent is paid for exactly what it is later judged on. An earlier version used a squared baseflow shortfall, $-\big((Q_k - Q_t)^+/Q_k\big)^2$, instead. A 15% shortfall then cost almost nothing in reward, but it zeroes the day in the score. PPO duly learned to shave baseflows to save water: its reward rose above the rule's while its functional-flow score fell from 0.85 to 0.70. We met the same trap in the stormwater-pond problem of the companion report. When the reward and the evaluation measure differ, RL optimizes the reward.

The pulse term pays the *increase* of the progress $\min(1, n_k/D_k)$ and never pays the same progress twice. A pulse therefore earns at most $\omega_k w_{ff}$ per year, even when the declared year type (and with it D_k) changes during the year. Partial pulses earn partial credit. Two loopholes in earlier versions are worth recording, because RL finds such holes quickly. Near-miss days were paid without counting toward the duration, so hovering just below the target paid more than completing the pulse. And the payment per day was w_p/D_k with the *current* D_k, so a change of year type mid-pulse paid twice. Scenarios combine the terms as follows:

Scenario	Hard river floor	Reward
1 MIF	minimum instream flow	$r^{\text{sup}} + r^{\text{fl}}$
2 Prescribed	prescribed FF hydrograph	$r^{\text{sup}} + r^{\text{fl}}$
3 FF reward	none	$r^{\text{sup}} + r^{\text{fl}} + r^{\text{ff}}$
4 Multi	none	$r^{\text{sup}} + r^{\text{fl}} + r^{\text{ff}} + r^{\text{hp}}$

The shortage is squared so that many small cuts are preferred to one big one, the usual reason for hedging. The weights are $w_s = 5$, $w_f = 1$, $w_{ff} = 5$ and $w_h = 0.1$. They are value judgements, not physics. A 15% shortage spread over the irrigation season costs about -16 per year, and raising the functional-flow score from 0 to 1 is worth $5\times$(number of components), i.e. 20–25 per year. Changing the weights traces out different points on the supply–ecology trade-off (Section 9).

To compare policies that were trained or operated under different scenarios, we use one common yardstick, J_{ff}: the mean per-year sum of $r^{\text{sup}} + r^{\text{fl}} + r^{\text{ff}}$, i.e. the Scenario 3 reward, computed for every policy.

4.5 Episodes, discounting and the end of an episode

An episode is three consecutive water years, starting from a random training year (WY1989–2016) and a random storage between 30% and 85% of capacity. Three years let carry-over storage matter, as in drought sequences. The discount factor is $\gamma = 0.995$, an effective horizon of about 200 days. That is long enough to see that water stored in winter pays off in summer.

The end of an episode is a *time limit*, not the end of the world. The reservoir would keep operating. Treating the last day as terminal would teach the agent that water left in storage at the end is worthless, and it would empty the reservoir. We therefore *bootstrap* at the time limit: the critic's estimate of the value of the final state is added to the last reward ($r_T \leftarrow r_T + \gamma\, V(s_{T+1})$). This addition to the training code (`train_ppo.py`, the `boot` entries) is the standard fix for time limits.

5 The simulator: one OHQ run per day

Every decision step builds and solves a one-day OHQ model through `openhydroqual_py` (`McClureBackend` in `ohq_eflows_env.py`):

- a `Reservoir` block with the day's storage and a constant inflow time series;
- three `fixed_head` receivers (River, Canal, Spill), whose storage accumulates the delivered volume, so day-mean flows are read exactly;
- two `Reservoir_link_rule` outlets with $Q_{\max}$ set to the action and a linear taper over the 1% of capacity above S_{in};
- a third `Reservoir_link_rule` as spillway, opening from S_{cap} to $1.01\, S_{\text{cap}}$ with a capacity of 5000 m^3/s;
- solver step 0.1 d, fixed (`max_timestep_increase_factor`=1), one solver thread per model (`n_threads`=1), because 24 models run in parallel.

The daily water balance $S_{t+1} = S_t + (I_t - R_t - C_t - W_t)\,\Delta t$ closes to about 10^{-4} hm^3, i.e. 10^{-7} of the storage. One worker runs about 150 simulated days per second. With 24 parallel workers training runs at about 1400 days per second, so four million simulated days (about 11,000 years of operation) take under an hour.

6 How the agent is trained

We use proximal policy optimization (PPO), the same actor–critic algorithm as in the companion report. An *actor* network maps o_t to the mean of a Gaussian over actions, and a *critic* network estimates the value $V(o_t)$. Training repeats: 24 workers each run the current policy for 512 days; advantages are computed with GAE ($\lambda = 0.95$); both networks are updated for 5 epochs with the clipped PPO objective (learning rate 5×10^{-5}, rewards scaled by the running standard deviation of the discounted return). Getting PPO to improve on the rule at all took several attempts. The lessons are as useful as the result:

1. **Learn a correction to the rule, not the rule itself (residual policy).** Our first attempt cloned the tuned prescriptive rule into the actor by regression and then ran PPO. The clone was never exact: its action errors of about ±6% in flow cost 5.7 of the rule's 7.85 reward points per year, and PPO only won part of that back. In the final set-up the agent's action is a *correction* added to the rule's action, $a_t = \mathrm{clip}\big(a^{\mathrm{rule}}(o_t) + \delta_\theta(o_t)\big)$, and the actor starts at $\delta_\theta \equiv 0$. Training therefore starts from the rule exactly. The correction has full authority, $\delta \in [-1, 1]$, so the agent can override the rule completely wherever that pays. This is "residual RL". The rule supplies the calendar of a functional-flow hydrograph, which is hard to learn from scratch, and RL adapts it to storage, forecasts and the passing hydrology.

2. **Exploration noise must not create penalties by itself.** PPO explores by adding Gaussian noise to the action. On a log-scale release, noise of standard deviation 0.018 is ±7% daily flow jitter. With the first, pass/fail version of the functional-flow reward, that jitter alone cost 8.6 points per year (the rule's whole reward was 7.6): mostly recession violations, because the prescribed recession ran at 95% of the limit. PPO then learned to cope with its own noise, which made the noise-free policy *worse*. The graded credit, the 7-day-mean recession and a recession at 80% of the limit cut the noise cost to 0.85 points per year at standard deviation 0.01, which is the exploration level used.

3. **Keep the best snapshot, judged on the real objective.** A checkpoint is saved every 18 updates. Each one is run *deterministically* (no exploration noise) on five training-period windows (starting 1990, 1996, 2002, 2008, 2012), and the one with the highest mean reward per year is kept. The test years are never used for any choice.

7 Baselines

Each scenario has a rule-based baseline, a *standard operating policy with hedging*:

- river: today's floor (MIF or prescribed hydrograph); in Scenarios 3–4, which have no floor, the prescribed hydrograph, so the baseline is the prescriptive operation scored with the functional-flow reward;
- flood control: storage above the rule curve $S_{\mathrm{toc}}(d)$ is released over n_{dd} days, up to Q_{safe};
- canal: full demand, cut to a fraction h_f of it when storage is below $h_S\, S_{\mathrm{cap}}$.

The three parameters (h_S, h_f, n_{dd}) were tuned for each scenario by grid search ($h_S \in \{0.2, 0.3, 0.4, 0.5\}$, $h_f \in \{0.5, 0.7, 0.85\}$, $n_{dd} \in \{5, 15\}$ d) on the same five training windows, maximizing that scenario's mean reward per year. The tuned baselines are therefore optimized for the same objectives as the agents.

8 Results

All results in this section are for the *test period*, water years 2017–2025, which no tuning or selection step has seen. It is run as three 3-year windows (WY2017–19, 2020–22, 2023–25), each started from the observed Lake McClure storage on October 1 and with the same forecast errors for every policy. The period is demanding: WY2017 and WY2023 are the two wettest years of the record, and WY2020–2022 is a three-year drought.

	Irrigation [%]		Flood	Energy	River	Functional-flow score				
Policy	mean	worst yr	d/yr	GWh/yr	hm^3/yr	all	dry	mod.	wet	J_{ff}
MIF rule	99.9	99.1	6.1	311	681	0.57	0.70	0.54	0.49	-1.0
Prescribed FF rule	97.3	85.0	7.0	319	715	0.93	0.95	0.95	0.91	6.5
Unimpaired flow (no dam)	–	–	–	–	–	0.88	0.88	0.92	0.87	–

Table 5: Test period WY2017–2025. Irrigation: delivered volume as % of demand, mean over the nine years and worst year. Flood: days with river flow above Q_{safe}. River: mean annual river flow below the dam (releases plus spill). Functional-flow scores by the May 1 declared year type. J_{ff}: mean per-year reward under the Scenario 3 weights (supply + flood + functional flows), the common yardstick. The last row scores the unimpaired inflow itself, as if there were no dam.

8.1 Summary of the operating modes

Table 5 and Figure 5 summarize the comparison. Read them as a trade-off between two goods: water delivered to the canal, and how closely the river below the dam follows a functional hydrograph.

- **Minimum instream flow (Scenario 1)** is almost free for irrigation: 99.9% of demand is delivered, and even the worst drought year gets 99%. Ecologically it is poor. The functional-flow score is 0.57 against 0.88 for the unimpaired river, and lowest in wet years (0.49). The licence flows supply no fall pulse, no spring pulse and no managed recession. Most of the score the river does get comes from uncontrolled spill in wet years, which is not a managed outcome.
- **Prescribed functional flows (Scenario 2)** lift the score to 0.93, above the unimpaired river, because the schedule is built to meet the median-based targets every year. The cost is concentrated in the drought: overall delivery drops only to 97.3%, but WY2022, the third dry year in a row, delivers 85% of demand. The extra river water averages about 34 hm^3/yr, roughly 5% of the irrigation demand, but it must be released in dry years too, when it hurts most.

Flood days (6–7 per year) are nearly the same for both: they come from WY2017 and WY2023, when inflows far exceeded the safe channel capacity whatever the operation. Energy differs little, since both outlets pass the same powerhouse and the total release is set mostly by the demand.

8.2 Year by year

Figure 6 breaks the comparison down by water year. Two patterns stand out. First, the functional-flow gain of the prescription is largest in wet and moderate years (2017–2019, 2023–2025). In those years the minimum-flow operation stores or diverts the spring snowmelt that a functional hydrograph would pass downstream. Second, the irrigation cost of the prescription appears only in WY2021 and WY2022, at the end of the drought. By then storage has fallen to about 250 hm^3 under the prescription, against about 330 hm^3 under minimum flows (Figure 8), and the hedging rule cuts deliveries. This is the classic problem with a static e-flow schedule. It spends water early in a drought without knowing that the drought will last, and the bill arrives two years later, in irrigation.

Table 6 shows which components carry the difference. Under minimum flows, the fall pulse (FA) and the spring pulse and recession (SP) are mostly missing, and the dry-season baseflow (DS) of wet

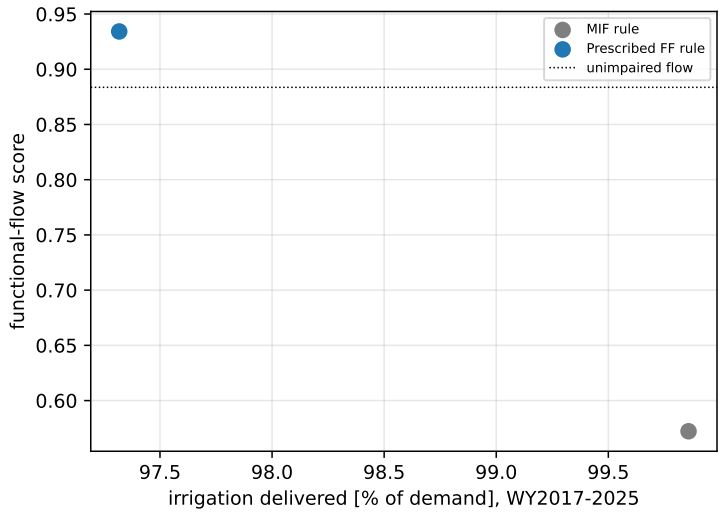


Figure 5: Irrigation delivered vs functional-flow score, WY2017–2025. Up and to the right is better. The dotted line is the score of the unimpaired river.

years is far below what the natural river would carry. The prescription meets almost everything. What it still misses comes from deciding under forecast uncertainty: a wet-season baseflow or peak flow sized for the year type declared in winter, in a year that the May 1 forecast then classes differently (Section 3.4).

8.3 Hydrographs

Figures 7–9 show what the operations look like day by day: river flow against the unimpaired inflow (log scale), storage, and canal delivery against demand.

8.4 A check against observed operations

The model is stylized, so it is worth asking whether its minimum-flow operation looks anything like the real river. Table 7 compares, for the years with a river record below the diversions (USGS near Snelling, WY1989–2009), the observed river volume and functional-flow score with those of the simulated minimum-flow rule. The observed river scores 0.53 on average and the simulated one 0.42. In dry and wet years the two agree closely (e.g. 1990, 1992, 1994, 2006, 2007). The gap is in moderate years. There the real river carried two to four times the licence minimum (e.g. 240–280 hm^3 against 70–150 hm^3 in 2002–2004) and scored 0.5–0.6 instead of 0.1–0.2. The extra water came from operational releases that the stylized model does not represent. The simulated minimum-flow rule is therefore a pessimistic stand-in for the status quo in moderate years, and a close one otherwise.

8.5 Reinforcement learning (Scenarios 3 and 4)

Result: in this set-up PPO did not improve on the tuned prescriptive rule. Starting from the rule exactly (residual policy, zero correction), every PPO update made the noise-free policy *worse* on the training-period validation windows (Figure 10, right). Checkpoint selection therefore returns update 0, i.e. the rule itself, and the "PPO" operation on the test period is identical to the

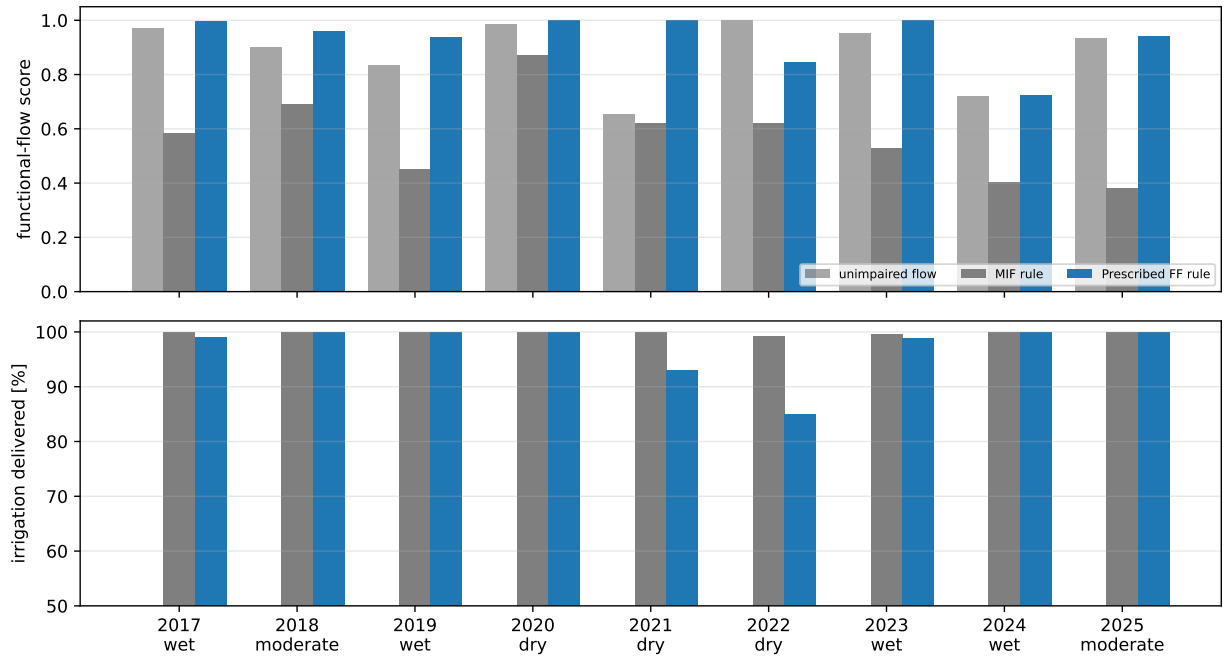


Figure 6: Functional-flow score (top) and irrigation delivered (bottom) by water year, with the year type declared on May 1.

prescribed functional-flow rule. It is not drawn separately in the figures above. Scenario 4 (adding hydropower) was not run, since Scenario 3 gave no learned policy to extend.

What PPO changed is clear from the reward components (Table 8). The functional-flow part stayed the same (score 0.85–0.86 throughout), but the policy released progressively more river water: up to 19 hm^3/yr more than the rule. That water was missing from irrigation in dry years, and the supply term fell from -4.5 to -7.8 per year. In other words, the agent paid irrigation water for no ecological gain.

Why. The most likely explanation is a bias that Gaussian exploration builds into this problem. During training every release carries random noise (here $\pm 4\%$). Near a target, a little extra release protects the *noisy* policy against falling short. The benefit is immediate, on the same day, and easy to attribute. The cost (less water in storage) appears months later, as a hedged irrigation delivery in late summer or in the next dry year. With $\gamma = 0.995$ and three-year episodes this delayed cost is real in the return, but it is hidden under much larger random variation between hydrologic years. PPO's advantage estimates pick up the immediate, noise-protecting benefit and largely miss the delayed cost. So the policy drifts toward over-releasing, which is the right reaction for the noisy policy and the wrong one for the deployed, noise-free policy. The training curve (left panel) shows no clear upward trend either. The signal that would distinguish a better adaptive rule from the prescription is small compared with the year-to-year noise.

What the attempts taught. The path to this result went through several failures, each instructive (Sections 4.4 and 6):

1. A cloned copy of a calendar-based rule is not the rule. Small action errors break day-specific targets, so we learned a correction to the rule (residual policy) instead.

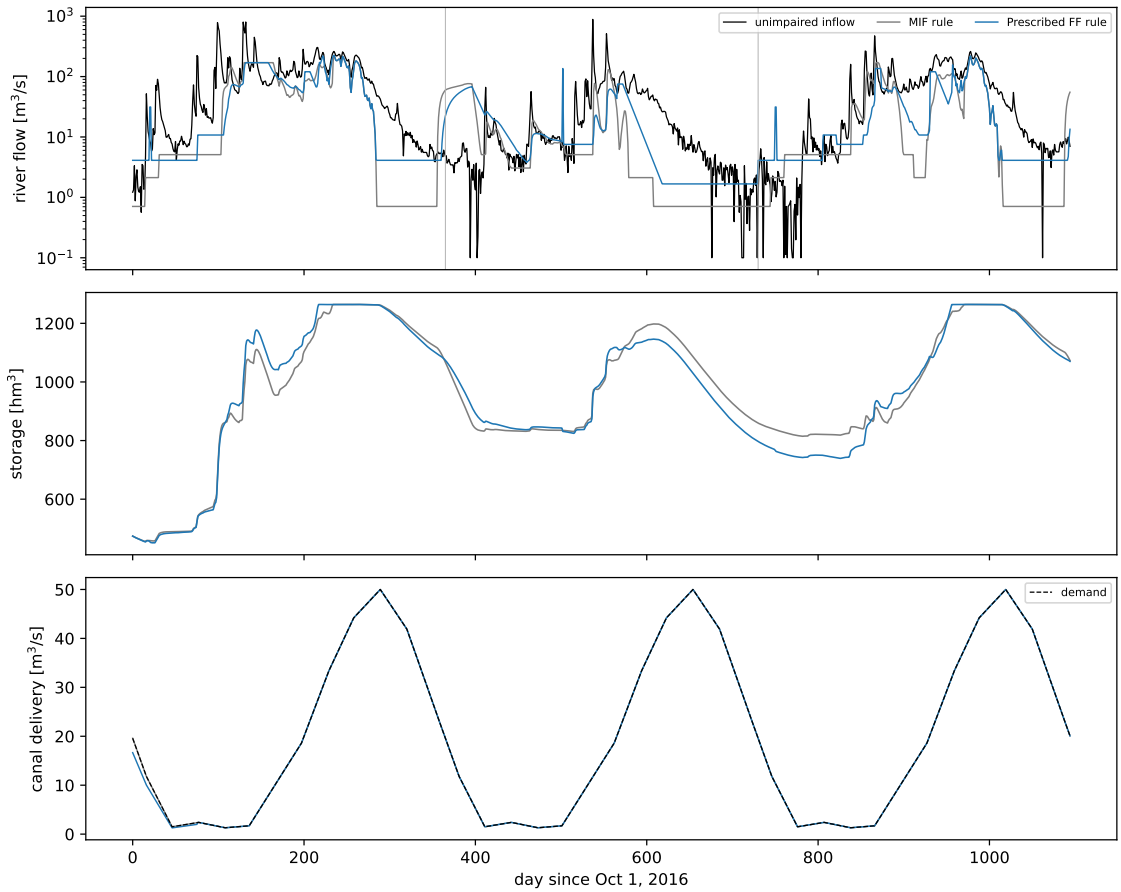


Figure 7: WY2017–2019 (wet, moderate, wet). The 2017 snowmelt fills the reservoir under both operations and forces spill, i.e. flood flows that no operation could avoid. Under the prescription the spring pulse and the gradual recession are visible in every year. Under minimum flows the river drops to the licence floor as soon as the spill ends.

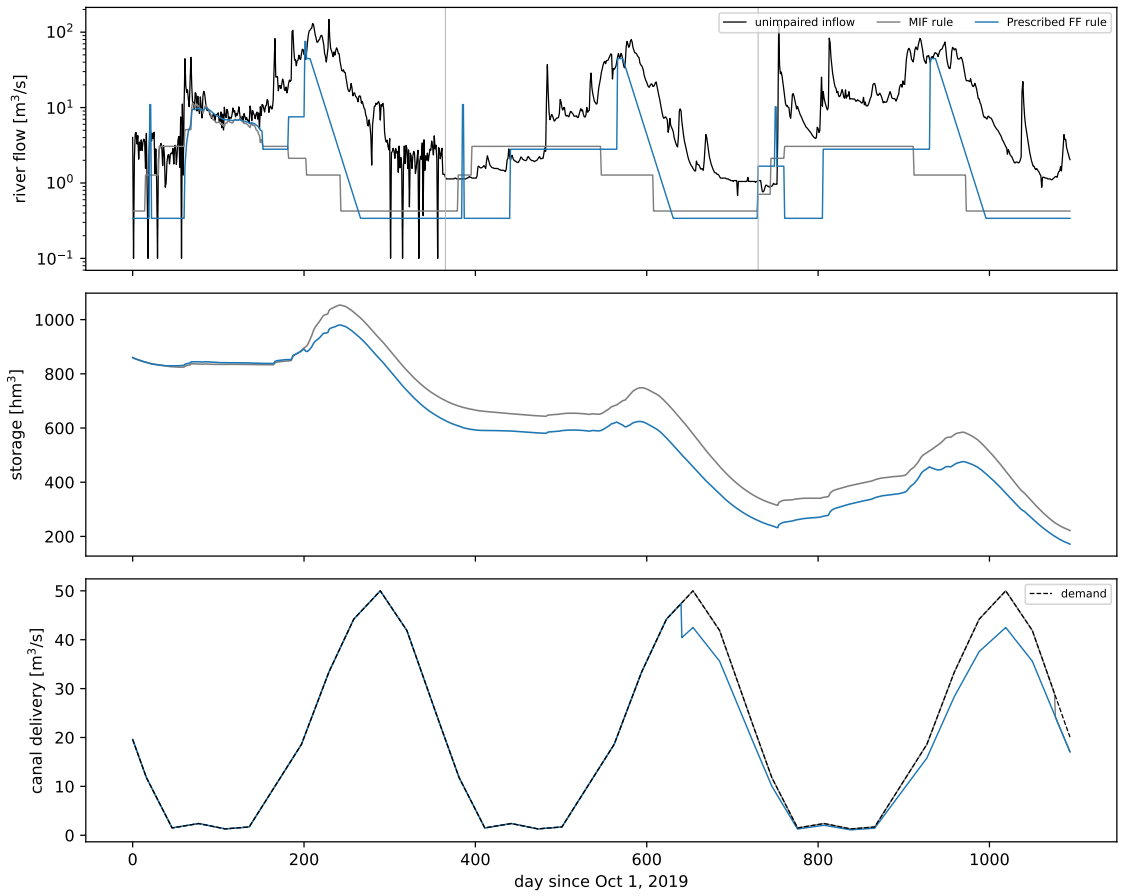


Figure 8: WY2020–2022, the drought. The prescription releases a spring pulse each year (top), and storage ends each year lower than under minimum flows (middle). In the third year this forces hedged canal deliveries (bottom): the irrigation cost of a static e-flow schedule in a multi-year drought.

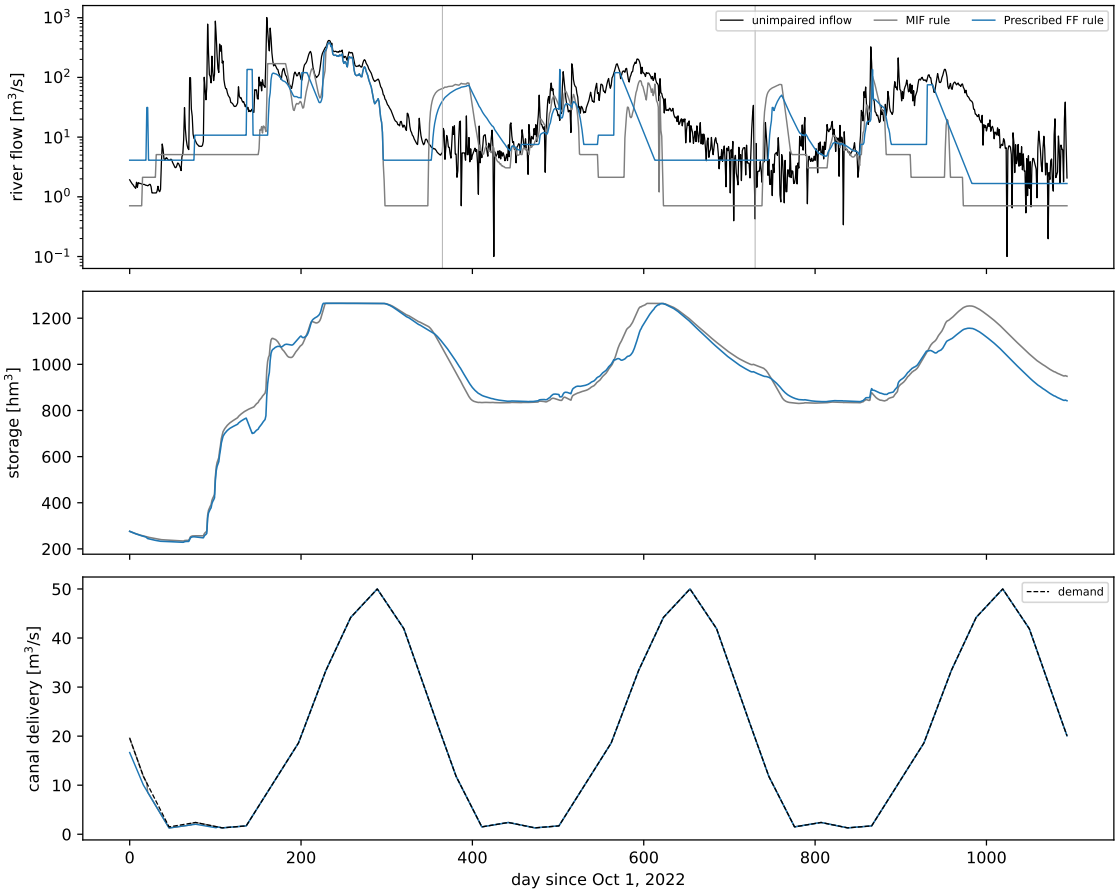


Figure 9: WY2023–2025 (wet, wet, moderate). After the 2020–22 drought the reservoir starts low, and the record 2023 runoff refills it within one season.

Policy	WY	type	Irr. [%]	Flood d	FA	WB	PK	SP	DS	FF
MIF rule	2017	wet	100.0	17	0.00	0.74	1.00	0.99	0.19	0.58
MIF rule	2018	moderate	100.0	0	1.00	0.43	1.00	0.90	0.13	0.69
MIF rule	2019	wet	100.0	7	0.00	0.56	0.99	0.54	0.17	0.45
MIF rule	2020	dry	100.0	0	1.00	1.00	–	0.48	1.00	0.87
MIF rule	2021	dry	100.0	0	0.00	1.00	–	0.48	1.00	0.62
MIF rule	2022	dry	99.1	0	0.00	1.00	–	0.48	1.00	0.62
MIF rule	2023	wet	99.6	31	0.00	0.28	1.00	0.99	0.37	0.53
MIF rule	2024	wet	100.0	0	1.00	0.45	0.00	0.44	0.13	0.41
MIF rule	2025	moderate	100.0	0	1.00	0.41	0.00	0.45	0.05	0.38
Prescribed FF rule	2017	wet	99.0	23	1.00	1.00	1.00	0.98	1.00	1.00
Prescribed FF rule	2018	moderate	100.0	0	1.00	0.81	1.00	1.00	1.00	0.96
Prescribed FF rule	2019	wet	100.0	7	1.00	0.71	1.00	0.98	1.00	0.94
Prescribed FF rule	2020	dry	100.0	0	1.00	1.00	–	1.00	1.00	1.00
Prescribed FF rule	2021	dry	93.0	0	1.00	1.00	–	1.00	1.00	1.00
Prescribed FF rule	2022	dry	85.0	0	0.39	1.00	–	1.00	1.00	0.85
Prescribed FF rule	2023	wet	98.9	33	1.00	1.00	1.00	1.00	1.00	1.00
Prescribed FF rule	2024	wet	100.0	0	1.00	0.48	0.14	1.00	1.00	0.72
Prescribed FF rule	2025	moderate	100.0	0	1.00	0.71	1.00	1.00	1.00	0.94

Table 6: Component scores per water year (FA fall pulse, WB wet-season baseflow, PK peak flow, SP spring pulse and recession, DS dry-season baseflow; "–" where a component does not apply).

2. RL found two reward loopholes within minutes of training: near-miss pulse credit that could be collected indefinitely, and double payment when the year type changed. Every reward term needs an explicit bound.
3. A reward that differs from the evaluation score gets optimized instead of the score. A squared baseflow penalty let PPO shave baseflows. The final reward is the daily increment of the score.
4. Pass/fail ecological thresholds on daily flows, and recession limits on raw daily changes, make the reward hypersensitive to the exploration noise that policy-gradient methods rely on.

9 Discussion and next steps

What this proof of concept shows

- **The problem is well posed and fast to simulate.** A daily OHQ reservoir model, driven by observed unimpaired inflow, runs at about 1400 simulated days per second on 24 cores, with mass balance closed to 10^{-7}. Functional flows can be scored automatically from CEFF-style targets derived from the same river's natural hydrology.
- **Prescribed functional flows work, at a drought cost.** A static functional-flow hydrograph scores well (0.93), but it spends storage early in a drought without knowing how long the drought will last. This is the opening an adaptive operation should exploit.
- **Off-the-shelf PPO does not exploit it here.** Even started exactly from the prescriptive rule, PPO drifted toward over-releasing. The extra release protects a noisy exploring policy, and its supply cost arrives months later. The improvement an adaptive rule could make is small compared with year-to-year hydrologic noise, so policy-gradient learning with per-day Gaussian exploration has a poor signal to work with.

		River volume [hm^3]		FF score	
WY	type	observed	MIF rule	observed	MIF rule
1989	moderate	123	72	0.18	0.10
1990	dry	110	53	0.58	0.62
1991	moderate	91	63	0.16	0.62
1992	dry	128	56	0.62	0.62
1993	wet	447	172	0.35	0.12
1994	dry	266	254	0.87	0.87
1995	wet	1343	1718	0.56	0.63
2002	moderate	276	67	0.61	0.10
2003	moderate	238	150	0.53	0.18
2004	moderate	237	72	0.62	0.10
2005	wet	943	826	0.50	0.27
2006	wet	1548	1474	0.79	0.80
2007	dry	330	119	0.87	0.87
2008	moderate	123	66	0.30	0.10
2009	moderate	177	78	0.49	0.10

Table 7: Observed river flow near Snelling vs the simulated minimum-flow operation (years with a complete record).

	reward/yr	supply	flood	FF reward	FF score	irrigation [%]	river [hm^3/yr]
tuned rule	4.53	-4.47	-0.37	9.38	0.846	95.6	395
PPO update 72	2.59	-6.32	-0.43	9.35	0.858	95.1	406
PPO update 144	2.18	-6.51	-0.66	9.34	0.857	95.1	407
PPO update 216	0.66	-7.85	-0.82	9.33	0.856	94.9	414
PPO update 324	1.85	-6.66	-0.74	9.25	0.853	95.1	409

Table 8: Validation windows (training period): reward components per year for the rule and for PPO checkpoints.

How to get an adaptive rule that beats the prescription

The results point to methods that evaluate a policy *without* exploration noise, on whole multi-year windows:

1. **Direct policy search.** Parameterize an adaptive rule with 10–50 parameters: e.g. pulse magnitudes and baseflow levels as functions of storage and the runoff forecast, or a small network. Tune the parameters with an evolution strategy (CMA-ES) on the deterministic multi-year objective. This is the evolutionary multi-objective direct policy search (EMODPS) used in reservoir operations. It needs no per-day credit assignment and gives Pareto fronts directly.

2. **Value-based RL with discrete releases** (the outline's DQN). Its exploration (occasional random actions) does not bias the policy in the same way. It is worth trying with a few release levels *relative to* the prescribed schedule.

3. **A larger signal.** The headroom grows with harder conditions: stricter targets, higher demand, or climate-perturbed inflows with earlier snowmelt. Under those conditions a static

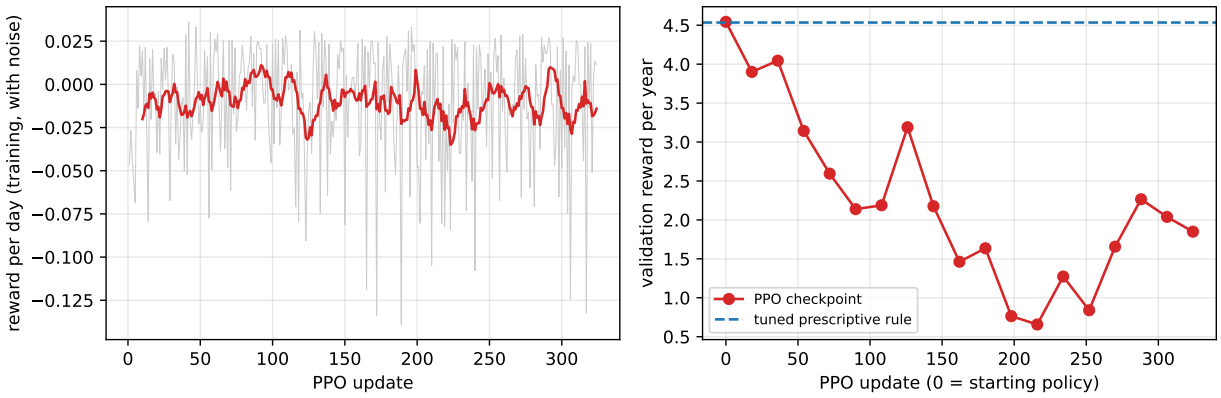


Figure 10: Scenario 3 training. Left: mean reward per simulated day of the noisy training policy (gray: per update; red: 10-update mean). Right: mean reward per year of each saved checkpoint, run without noise on the five training-period validation windows, against the tuned prescriptive rule (dashed). Update 0 is the rule with zero correction.

schedule fails more often, and adaptivity is worth more.

Simplifications to keep in mind

- The system is McClure-*like*: one reservoir, no re-regulating reservoirs, no evaporation, no groundwater, approximate head and flood rule. The FERC/Davis–Grunsky flows are stylized.
- Functional-flow windows are fixed calendar windows. CEFF times the components from the hydrograph, and an agent could be rewarded for timing too.
- The demand is the same every year (mean observed diversions); real demand is higher in dry years.
- Water temperature is not modeled. Cold-water pool management is a natural extension in OHQ, which already carries heat and constituents.

Next steps suggested by this proof of concept

1. **Trade-off curves.** Repeat Scenario 3 for several w_{ff} and plot irrigation reliability against FF score. This is a Pareto front of adaptive operations to set against the single point of a prescription.
2. **Comparison with the real system.** Compare with CenSierraPywr and with historical operations, using standard reservoir performance metrics (reliability, resilience, vulnerability) by water-year class.
3. **Deep uncertainty.** Train on historical years, test on climate-perturbed inflows (warming shifts snowmelt earlier). This is where adaptive rules should beat static ones most.
4. **Algorithm.** The outline plans DQN (discrete releases). PPO with continuous actions is used here because it handles the two-dimensional, wide-range release decision directly. A DQN version with a few discrete release levels is straightforward to add for comparison.

5. **Interpretability.** Distill the trained network into a readable rule (e.g. a shallow decision tree on storage, forecast and season), so that it can be written into a licence.

Reproducing

```
python prepare_merced.py            # download CDEC/USGS data
python plot_eflows_setup.py         # setup figures and tables
python evaluate_eflows.py --tune    # tune the rule baselines
python train_ppo.py --env eflows \
  --env-kwargs '{"scenario": "ffreward", "residual": true}' \
  --total-steps 3981312 --num-envs 24 --rollout-steps 512 \
  --gamma 0.995 --lr 5e-5 --epochs 5 --normalize-rewards \
  --reward-clip 0 --save-every 18 --bc-rule ohq_eflows_env:zero_policy \
  --bc-log-std -4.6 --out results/ppo_eflows_ffreward.pt
python evaluate_eflows.py --select "results/ppo_eflows_ffreward_u*.pt" \
  --scenario ffreward --env-kwargs '{"residual": true}'
python evaluate_eflows.py --compare # test-period tables and figures
```

References

- Yarnell, S.M., et al. (2020). A functional flows approach to selecting ecologically relevant flow metrics for environmental flow applications. *River Research and Applications* 36(2), 318–324; and the California Environmental Flows Framework (CEFF), California Environmental Flows Working Group (2021).

- *Shifting trade-offs: finding the sustainable nexus of hydropower and environmental flows in the San Joaquin River watershed* (2022), Frontiers; the functional-flow implementation (its Section 2.3) that the e-flow paper plans to replace with an RL agent.

- Schulman, J., et al. (2017). Proximal policy optimization algorithms. arXiv:1707.06347.